

Stochastic Lie-Bracket Approximations for Zeroth-Order Optimization on Manifolds

Mahmoud Abdelgalil, Miroslav Krstic, and Jorge I. Poveda

Abstract—We propose an algorithm for zeroth-order global optimization of smooth functions on compact manifolds. The algorithm is expressed as an ordinary differential equation driven by piecewise-smooth McShane-type approximations of the Wiener process, and requires only zeroth-order (measurement) access to the objective. We show that the trajectories of this ODE approximate, in the mean-square sense, a Riemannian Langevin diffusion whose unique invariant Gibbs measure concentrates on the set of global minimizers of the objective. By exploiting this stochastic convergence-of-trajectories property, we establish probabilistic guarantees of convergence for the algorithm. We then analyze the long-time behavior of the algorithm directly, leading to infinite-horizon practical probabilistic guarantees.

Index Terms—Stochastic optimization, Zeroth-Order Optimization, Optimization on Manifolds.

I. INTRODUCTION

MODEL-FREE optimization algorithms that rely only on real-time measurements of an objective function play an important role in adaptive control, source seeking, and learning in dynamical systems. A prominent approach is extremum seeking (ES) [1], which combines persistent probing with averaging techniques to provide control-theoretic stability guarantees. In particular, Lie-bracket-based interpretations of ES [2] have enabled systematic designs with bounded update rates [3], local asymptotic stability [4], and global practical stability [5], with extensions to systems evolving on manifolds [4], [6]–[8] and hybrid systems [9]–[11]. Nevertheless, smooth deterministic ES on manifolds inherits a fundamental limitation of the gradient flows it approximates: convergence is generally local or, at best, almost global. Hybrid ES based on synergistic switching can overcome this limitation [10], [12], but typically requires additional information about the cost function to escape undesirable equilibria.

A classical approach to overcoming spurious equilibria in nonconvex Euclidean optimization is to inject persistent stochasticity, transforming the gradient flow into a Langevin

diffusion whose invariant Gibbs measure concentrates near global minimizers. This principle underlies continuous-time simulated annealing [13]–[15] and stochastic approximation [16], [17], and has also motivated Langevin-based methods on Riemannian manifolds [18]. However, these approaches typically rely on gradient information and are therefore not directly applicable to the zeroth-order setting.

Motivated by this background, in this paper we introduce a novel *stochastic Lie-bracket approximation* (SLBA) algorithm that combines the bounded-update-rate design of [3] with Brownian, rather than periodic, dithering. The algorithm is modeled by a *random ODE* on a smooth compact manifold $\mathcal{M}$, driven by piecewise-smooth McShane-type approximations of a Wiener process [19]. In the Wong–Zakai limit [20]–[23], its trajectories converge to those of a Stratonovich SDE whose generator, under a suitable drift compensation, coincides with that of a Riemannian Langevin diffusion, despite the algorithm using only zeroth-order information. The resulting Gibbs invariant measure concentrates on the global minimizers of the objective function. We quantify this concentration and establish finite-horizon practical guarantees through mixing properties of the limiting SDE.

We further study the algorithm itself for a fixed discretization parameter $\varepsilon > 0$. Its sampled dynamics define a Markov chain for which we establish the existence of invariant measures and characterize their long-time behavior, showing weak convergence toward the Gibbs measure as $\varepsilon \rightarrow 0$. This yields global practical probabilistic optimization guarantees over infinite horizons and from arbitrary initial conditions. Compared with deterministic Lie-bracket ES on manifolds [6], Brownian dithering replaces local practical convergence with global measure-theoretic guarantees. Relative to stochastic (first-order averaging) ES [24], the main novelties are the incorporation of the manifold setting via Lie brackets and, importantly, the invariant-measure analysis of the approximating dynamics at fixed ε . This Markov-chain perspective is related to the analysis of numerical SDE schemes [25], [26], although our algorithm is instead a continuous-time random ODE.

The remainder of the paper is organized as follows: Section II presents the notation, Section III describes the algorithm and the trajectory approximation property, Section IV studies the limiting SDE and its finite-horizon analysis, Section V presents the infinite-horizon and invariant-measure analysis, the numerical results are presented in Section VII, and the conclusions are presented in Section VIII.

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II. NOTATION AND PRELIMINARIES

We denote the natural numbers by $\mathbb{N}_0 := \{0, 1, 2, \dots\}$ and $\lceil a \rceil$ is the least integer bigger than or equal to a . For a subset A of a metric space, $\bar{A}$ and $\text{int}(A)$ denote closure and interior, 1_A the indicator function, and, for $\eta > 0$ let $A^\eta := \{y : d(y, A) < \eta\}$ be the open η -neighbourhood of A . Lebesgue measure on $\mathbb{R}^N$ is written Leb , and $B_\rho(a) \subset \mathbb{R}^N$ is the open Euclidean ball of radius ρ centred at a . The Euclidean norm is denoted by $|\cdot|$ and $\|\cdot\|$ the induced operator norm. For a linear map L , $\sigma_{\min}(L)$ is its smallest singular value. Throughout the manuscript, $\mathcal{M}$ denotes a smooth, connected, compact, orientable n -dimensional manifold without boundary, embedded in $\mathbb{R}^m$, $T_x\mathcal{M}$ and $T_x^*\mathcal{M}$ are its tangent and cotangent spaces at x , respectively, and $\mathcal{B}(\mathcal{M})$ denotes its Borel σ -algebra. We use $C^\infty(\mathcal{M})$, $C(\mathcal{M})$, and $\mathfrak{X}(\mathcal{M})$ for smooth functions, continuous functions, and smooth vector fields on $\mathcal{M}$, respectively, and $\text{span}_{\mathbb{R}}$ for the linear span with constant real coefficients. For $f, h \in \mathfrak{X}(\mathcal{M})$ and $\psi \in C^\infty(\mathcal{M})$, $f(\psi)$ denotes the derivative of ψ along f , $[f, h]$ the Lie bracket, $f \otimes h$ is the tensor product, and Φ_s^f is the time- s flow of f . Differentials of maps are written D , and are regarded as maps into the ambient $\mathbb{R}^m$ whenever differentials at distinct base points are compared. Given a Riemannian metric $\langle \cdot, \cdot \rangle_g$ on $\mathcal{M}$, the symbols grad , div , Δ , Hess , ∇ , and vol denote the gradient, divergence, Laplace–Beltrami operator, Hessian, Levi-Civita connection, and Riemannian density, respectively, and $\text{Vol}(\mathcal{M}) := \int_{\mathcal{M}} \text{vol}$. We write $d(\cdot, \cdot)$ for the geodesic distance and $B(z, \rho) \subset \mathcal{M}$ for the open geodesic ball. Since $\mathcal{M}$ is compact, d and the ambient distance $|x - y|$ are equivalent, modulo constants. We use $\|\psi\|_\infty := \sup_{\mathcal{M}} |\psi|$ to denote the supremum norm, and $\text{Lip}(\psi)$ to denote the smallest constant L with $|\psi(x) - \psi(y)| \leq L d(x, y)$. For a positive measure ν on $\mathcal{M}$, we equip the space $L^p(\nu)$ with the norm $\|\psi\|_{L^p(\nu)} := (\int_{\mathcal{M}} |\psi|^p d\nu)^{1/p}$. We use $\mathcal{P}(\mathcal{M})$ to denote the Borel probability measures on $\mathcal{M}$ with the topology of weak convergence, and $\rightharpoonup$ to denote weak convergence. For $\nu, \nu' \in \mathcal{P}(\mathcal{M})$, we use $\|\nu - \nu'\|_{\text{TV}}$ to denote the total variation distance and

$$d_{\text{BL}}(\nu, \nu') := \sup \left\{ \int_{\mathcal{M}} f d(\nu - \nu') : \|f\|_\infty \leq 1, \text{Lip}(f) \leq 1 \right\}, \quad (1)$$

to denote the bounded-Lipschitz distance, which metrizes weak convergence on the compact space $\mathcal{M}$. We use $\mathcal{N}(0, \Sigma)$ to denote the centered Gaussian distribution with covariance Σ . Finally, we use $\mathbb{E}$ and $\mathbb{P}$ to denote expectation and probability, respectively.

III. PROBLEM FORMULATION AND THE SLBA ALGORITHM

Let $\phi \in C^\infty(\mathcal{M})$ be the objective function, accessible only through evaluations $\phi(x)$. Our goal is to solve the problem

$$\min_{x \in \mathcal{M}} \phi(x). \quad (2)$$

Let $\phi_\star := \min_{x \in \mathcal{M}} \phi(x)$ and define the set of minimizers

$$S_\star := \{x \in \mathcal{M} : \phi(x) = \phi_\star\}.$$

To solve problem (2), we will design an algorithm expressed as a *random ODE*, whose trajectories converge to the set $S_\star$.

To that end, let $\mathcal{F} = \{f_1, \dots, f_r\} \subset \mathfrak{X}(\mathcal{M})$ be a family of vector fields satisfying the span condition

$$\text{span}_{\mathbb{R}}\{f_1(x), \dots, f_r(x)\} = T_x\mathcal{M}, \quad \forall x \in \mathcal{M}. \quad (3)$$

Such a family always exists, e.g., the tangential projections of the ambient coordinate vector fields of the embedding $\mathcal{M} \subset \mathbb{R}^m$ provide $r = m$ fields satisfying (3). Let $\mathcal{W} := \{W \in C([0, \infty); \mathbb{R}^{2r}) : W_0 = 0\}$ be the standard Wiener space, i.e., the space of continuous functions equipped with the Wiener measure, and, for any $W \in \mathcal{W}$, let

$$w_k^\ell := W_{(k+1)\varepsilon}^\ell - W_{k\varepsilon}^\ell$$

denote the increments over the grid of step $\varepsilon > 0$, with $k \in \mathbb{N}_0$, and $\ell \in \{1, \dots, 2r\}$. For $\kappa > 0$, define the functions

$$U_\kappa^1(s) := s + \frac{1 + \sqrt{2\pi^2\kappa + 1}}{2\pi} \sin(2\pi s), \quad (4)$$

$$U_\kappa^2(s) := s + \frac{1 + \sqrt{2\pi^2\kappa + 1}}{\pi} \sin^2(\pi s), \quad (5)$$

and, for $t \in [k\varepsilon, (k+1)\varepsilon)$, $i \in \{1, \dots, r\}$, $j \in \{1, 2\}$, the piecewise-smooth process

$$B_\varepsilon^{2i+j-2}(t, W) := W_{k\varepsilon}^{2i+j-2} + w_k^{2i+j-2} U_\kappa^{i_j}(w_k) \left(\frac{t - k\varepsilon}{\varepsilon} \right), \quad (6)$$

where the random index $i_j^j(w)$ is defined by

$$i_j^j(w) := \begin{cases} j & w^{2i-1} w^{2i} \geq 0 \\ 3-j & w^{2i-1} w^{2i} < 0 \end{cases}$$

i.e., the two functions are swapped within the i -th pair when the corresponding increments have opposite signs. This piecewise smooth interpolation of Brownian motion is known as a McShane-type interpolation [19]. By construction, B_ε^ℓ is piecewise differentiable, and it has a well-defined piecewise-continuous derivative $b_\varepsilon^\ell := \dot{B}_\varepsilon^\ell$. The Stochastic Lie-Bracket Approximation (SLBA) algorithm is then defined by the following random ODE:

$$\dot{x} = \bar{f}_0(x) + \frac{1}{\sqrt{\kappa}} \sum_{i=1}^r \sum_{j=1}^2 F_j(\phi(x)) f_i(x) b_\varepsilon^{2i+j-2}(t, W), \quad (7)$$

with $F_1 = \cos$, $F_2 = -\sin$, and a compensation drift $\bar{f}_0 \in \mathfrak{X}(\mathcal{M})$, specified later by Theorem 2 in terms of the family of vector fields $\mathcal{F}$ only, and independently from the function ϕ . Note that (7) is an ODE for each sample path W , and the objective enters only through the measured values $\phi(x(t))$. For each initial condition $x_0 \in \mathcal{M}$ and each sample path W , we write $x_t^\varepsilon(x_0)$ to denote the solution of (7). Solutions exist globally and remain on $\mathcal{M}$ since the right-hand side is tangent to $\mathcal{M}$.

By the Wong–Zakai theory for approximations with corrections [19], [21, Ch. 7], the trajectories of (7) approximate, as $\varepsilon \rightarrow 0$, the sample paths of the following Stratonovich SDE:

$$\begin{aligned} d\bar{x}_t = & \left(\bar{f}_0(\bar{x}) + \sum_{i=1}^r [F_1(\phi) f_i, F_2(\phi) f_i](\bar{x}) \right) dt \\ & + \frac{1}{\sqrt{\kappa}} \sum_{i=1}^r \sum_{j=1}^2 F_j(\phi(\bar{x})) f_i(\bar{x}) \bullet dW_t^{2i+j-2}, \end{aligned} \quad (8)$$

where the bracket term is the drift correction generated by (4)–(6). We write $\bar{x}_t(x_0)$ for the solution of (8) and $\bar{P}_t$ for its Markov semigroup. The following theorem, adapted from the results of [21, Thm. 7.2 and Rem. 7.2], establishes a *stochastic convergence of trajectories* property that will be instrumental for our result:

Theorem 1: *For every $\delta, T > 0$, there exists $\varepsilon^*(\delta, T) > 0$ such that for all $\varepsilon \in (0, \varepsilon^*)$,*

$$\sup_{x_0 \in \mathcal{M}} \mathbb{E} \left[\sup_{t \in [0, T]} \|x_t^\varepsilon(x_0) - \bar{x}_t(x_0)\|^2 \right] < \delta, \quad (9)$$

when both solutions are driven by the same sample path. $\square$

In what remains of this manuscript, we utilize the stochastic convergence of trajectories property established by Theorem 1, in conjunction with the properties of the limiting SDE (8), to analyze the behavior of the random ODE (7).

IV. THE LIMITING DIFFUSION AND ITS GIBBS MEASURE

We begin by analyzing the limiting SDE (8). The span condition (3) induces a canonical metric on $\mathcal{M}$. Indeed, since the co-metric $\sum_{i=1}^r f_i \otimes f_i$ is positive definite by virtue of (3), there exists a unique Riemannian metric $\langle \cdot, \cdot \rangle$ on $\mathcal{M}$ such that, $\forall \psi \in C^\infty(\mathcal{M})$,

$$\text{grad } \psi = \sum_{i=1}^r f_i(\psi) f_i. \quad (10)$$

In the sequel, all geometric operators refer to this metric.

Theorem 2: *Let $\langle \cdot, \cdot \rangle$ be the metric (10) and define*

$$\bar{f}_0 = -\frac{1}{2\kappa} \sum_{i=1}^r \nabla_{f_i} f_i.$$

Then, the following holds:

(a) *The infinitesimal generator of the SDE (8) is given by*

$$\mathcal{L}\psi = \frac{1}{2\kappa} \Delta\psi - \langle \text{grad } \phi, \text{grad } \psi \rangle_g, \quad (11)$$

for all $\psi \in C^\infty(\mathcal{M})$.

(b) *The SDE (8) admits the unique invariant measure*

$$\mu_\kappa = c_\kappa^{-1} e^{-2\kappa\phi} \text{vol}, \quad (12)$$

where $c_\kappa := \int_{\mathcal{M}} e^{-2\kappa\phi} \text{vol}$. $\square$

Remark 1: *A key implication of Theorem 2 is that the infinitesimal generator of the SDE (8) coincides with the generator of the Riemannian Langevin SDE defined by*

$$dx = -\text{grad } \phi(x) dt + \kappa^{-1/2} dB^g, \quad (13)$$

where B^g is the Brownian motion on $\mathcal{M}$, with respect to the metric g , whose generator is Δ [21, Chapter V]. This relationship plays a key role in our analysis. $\square$

The relationship between the SDE (8) and the Langevin dynamics (13) establishes (8) as a suitable optimization algorithm. In particular, by leveraging the explicit form of the invariant measure μ_κ , provided by Theorem 2, we can study its asymptotic behavior when $\kappa \gg 1$. To that end, we have the following proposition. We recall that $S_\star$ denotes the set of minimizers of ϕ .

Proposition 1: *For every closed $K \subseteq \mathcal{M}$ with $K \cap S_\star = \emptyset$ there exist $C_K, c_K > 0$, such that*

$$\mu_\kappa(K) \leq C_K e^{-2\kappa c_K}, \quad (14)$$

for all $\kappa > 0$. In particular, $\mu_\kappa(U) \rightarrow 1$ as $\kappa \rightarrow \infty$ for every open $U \supset S_\star$. $\square$

Although the invariant measure μ_κ concentrates on the set of minimizers $S_\star$ in the limit $\kappa \rightarrow \infty$, the convergence of the marginals of the process (8) to the invariant measure μ_κ is, in general, slow for large κ . This is due to the fact that, for a general compact boundary-less manifold $\mathcal{M}$ and a general function ϕ , the spectral gap of the operator $\mathcal{L}$ may decay in the limit $\kappa \rightarrow \infty$. Nevertheless, for a fixed $\kappa > 0$, the spectral gap of $\mathcal{L}^*$ is uniformly bounded away from zero and, as a result, convergence is guaranteed. On the other hand, there is an important case wherein the spectral gap of $\mathcal{L}^*$ can be uniformly upper bounded away from 0 in the limit $\kappa \rightarrow \infty$. The following proposition formalizes these claims.

Proposition 2: *For any $\kappa > 0$, let ρ_κ be the density of the invariant measure μ_κ given by (12). Then, the following holds:*

(a) *There exists $\lambda_\kappa > 0$ such that for any initial probability density $\rho_0 \in C^\infty(\mathcal{M})$ the solution ρ_t to the Fokker-Planck equation*

$$\partial_t \rho_t = \mathcal{L}^* \rho_t = \text{div}(\rho_t \text{grad } \phi) + \frac{1}{2\kappa} \Delta \rho_t, \quad (15)$$

satisfies

$$\|\rho_t - \rho_\kappa\|_{L^1(\text{vol})} \leq e^{-\lambda_\kappa t} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})}, \quad (16)$$

for all $t > 0$.

(b) *If $\phi \in C^\infty(\mathcal{M})$ is a Morse function with a unique local minimum, i.e., there exists a unique critical point $x_\star \in \mathcal{M}$ of ϕ for which $\text{Hess } \phi(x_\star) \succ 0$, then there exists $\lambda > 0$ such that (16) holds uniformly for all $\kappa > 0$.* $\square$

Having characterized the invariant measure of the limiting Riemannian Langevin diffusion, we next quantify the rate at which the distribution of its solutions approaches this measure. The following result establishes exponential convergence in total variation, with a rate determined by the spectral gap of the infinitesimal generator.

Proposition 3: *For any $\kappa > 0$, let $\lambda_\kappa > 0$ be the spectral gap of $\mathcal{L}$. Let $\bar{x}_t$ be a solution of the SDE (8) such that the initial condition $\bar{x}_0 \sim \rho_0 \text{vol}$, for any smooth probability distribution ρ_0 . Then, for every measurable set $U \subset \mathcal{M}$, we have that*

$$|\mathbb{P}\{\bar{x}_t \in U\} - \mu_\kappa(U)| \leq \frac{1}{2} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})} e^{-\lambda_\kappa t}, \quad (17)$$

for all $t \geq 0$. In particular, if $\rho_0 = (\int_{\mathcal{M}} \text{vol})^{-1}$, i.e. the uniform probability density, then

$$|\mathbb{P}\{\bar{x}_t \in U\} - \mu_\kappa(U)| \leq \frac{1}{2} e^{\kappa\gamma_\phi - \lambda_\kappa t}, \quad (18)$$

where $\gamma_\phi := \max_{x \in \mathcal{M}} \phi(x) - \min_{x \in \mathcal{M}} \phi(x)$. $\square$

Based on the properties we have obtained so far for the limiting SDE (8), we may now utilize the stochastic convergence-of-trajectories property established by Theorem

1. In particular, if x_t^ε and $\bar{x}_t$ are the solutions for the random ODE (7) and the SDE (8), respectively, from the same initial condition $x_0^\varepsilon = \bar{x}_0 \sim \rho_0 \text{ vol}$ and for the same driving Brownian motion path W , then the bound in Theorem 1 implies that the probability measure of x_t converges to the probability measure of $\bar{x}_t$ in the limit $\varepsilon \rightarrow 0$, uniformly on any compact interval $[0, T]$. In particular, we have the following proposition.

Proposition 4: *For any $\kappa > 0$, let $\lambda_\kappa > 0$ be the spectral gap of $\mathcal{L}$. Then, for any $T > 0$, any $\varepsilon > 0$, any $\kappa > 0$, and any open set $U \subset \mathcal{M}$ with a smooth boundary, there exists $\varepsilon^* \in (0, \infty)$ such that the following bound holds*

$$|\mathbb{P}\{x_t^\varepsilon \in U\} - \mu_\kappa(U)| < \frac{1}{2} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})} e^{-\lambda_\kappa t} + \varepsilon.$$

for all $\varepsilon \in (0, \varepsilon^*)$ and all $t \in [0, T]$. $\square$

V. INVARIANT MEASURES OF THE ALGORITHM

The exponential bound established in Proposition 4 is restricted to compact time intervals and therefore does not characterize the asymptotic behavior of the system as $t \rightarrow \infty$. In this section, we remove that limitation by directly analyzing the limiting behavior of the probability measure of (7). Throughout, the constants $\varepsilon, \kappa > 0$ are fixed unless stated otherwise. Also, for a Markov transition kernel $P : \mathcal{M} \times \mathcal{M} \rightarrow [0, 1]$ on $\mathcal{M}$, we write $P\psi(x) := \int_{\mathcal{M}} \psi(y) P(x, dy)$, and P^k for the k -step kernel.

Let $t = k\varepsilon$ and define the sampled process $X_k := x_{k\varepsilon}^\varepsilon$, where x_t^ε is a solution to (7). Then, we have the following result.

Lemma 1: *There exists a measurable map $\Psi_\varepsilon : \mathcal{M} \times \mathbb{R}^{2r} \rightarrow \mathcal{M}$ such that the sampled process X_k satisfies*

$$X_{k+1} = \Psi_\varepsilon(X_k, w_k),$$

where the increments w_k are i.i.d. samples from $\mathcal{N}(0, \varepsilon I_{2r})$. That is, (X_k) is a time-homogeneous Markov chain with the transition kernel

$$P_\varepsilon(x, A) = \mathbb{P}\{\Psi_\varepsilon(x, w) \in A\}.$$

Moreover, $P_\varepsilon\psi \in C(\mathcal{M})$ for any $\psi \in C(\mathcal{M})$, where

$$P_\varepsilon\psi(x) := \int_{\mathbb{R}^{2r}} \psi(\Psi_\varepsilon(x, w)) \gamma_\varepsilon(dw), \quad \gamma_\varepsilon := \mathcal{N}(0, \varepsilon I_{2r}).$$

That is, P_ε maps $C(\mathcal{M})$ into $C(\mathcal{M})$. $\square$

The Markovian characterization of the sampled dynamics in Lemma 1 enables us to study their long-run behavior through the associated transition kernel P_ε . In particular, the Feller property established above, together with the compactness of $\mathcal{M}$, guarantees the existence of an invariant probability measure for every fixed $\varepsilon > 0$, as formalized next.

Proposition 5: *For every $\varepsilon > 0$, the chain (X_k) admits an invariant probability measure $\pi_\varepsilon \in \mathcal{P}(\mathcal{M})$.* $\square$

Having established the existence of an invariant measure, it is natural to study the long-time behavior of the chain. The following result characterizes such behavior (cf. (1)):

Theorem 3: *Fix $\kappa > 0$ and let μ_κ be the unique invariant measure (12). Then, for every $\delta > 0$, there exist $T > 0$ and*

$\varepsilon^* > 0$ such that, for all $\varepsilon \in (0, \varepsilon^*)$ and all integers $k \geq T/\varepsilon$, we have that

$$\sup_{x \in \mathcal{M}} d_{\text{BL}}(P_\varepsilon^k(x, \cdot), \mu_\kappa) \leq \delta.$$

In particular, every invariant probability measure μ_ε of P_ε satisfies $d_{\text{BL}}(\mu_\varepsilon, \mu_\kappa) \leq \delta$ for all $\varepsilon \in (0, \varepsilon^*)$. $\square$

Theorem 3 establishes uniform convergence of the transition probabilities of the approximating dynamics toward the Gibbs measure μ_κ in the bounded-Lipschitz metric. We next translate this measure-theoretic approximation into explicit probabilistic bounds relative to the set $S_\star$ of global minimizers. In particular, the following lemma provides lower bounds on the probability of being in any neighborhood of $S_\star$, as well as upper bounds on the probability of being in closed sets separated from $S_\star$.

Lemma 2: *In the setting of Theorem 3, for all $\varepsilon \in (0, \varepsilon^*)$ and all integers $k \geq T\varepsilon^{-1}$, the following holds:*

- 1) *If $U \subsetneq \mathcal{M}$ is an open neighborhood of $S_\star$ and $\eta_U := \min\{1, \frac{1}{2}d(S_\star, \mathcal{M} \setminus U)\}$, then*

$$\inf_{x \in \mathcal{M}} P_\varepsilon^k(x, U) \geq 1 - C_U e^{-2\kappa C_U} - \delta \eta_U^{-1};$$

- 2) *If $\emptyset \neq K \subseteq \mathcal{M}$ is closed with $K \cap S_\star = \emptyset$ and $\eta_K := \min\{1, \frac{1}{2}d(K, S_\star)\}$, then*

$$\sup_{x \in \mathcal{M}} P_\varepsilon^k(x, K) \leq C_K e^{-2\kappa C_K} + \delta \eta_K^{-1},$$

where $C_U, c_U > 0$ are the constants of (14) associated with the closed set $\mathcal{M} \setminus S_\star^{\eta_U}$, and $C_K, c_K > 0$ are those associated with $\overline{K^{\eta_K}}$. In particular, they depend only on U , respectively K , and are independent of κ, ε, k , and x . $\square$

VI. PROOFS

Proof of Theorem 2: To simplify notation, let $\phi_j = F_j \circ \phi$. Utilizing the properties of the Lie Bracket, we compute

$$[\phi_1 f_i, \phi_2 f_i] = (\phi_1 f_i(\phi_2) - \phi_2 f_i(\phi_1)) f_i. \quad (19)$$

Consequently, the limiting SDE (8) simplifies to

$$d\bar{x}_t = f_0(\bar{x}) dt + \frac{1}{\sqrt{\kappa}} \sum_{i=1}^r \sum_{j=1}^2 \phi_j f_i(\bar{x}) \bullet dW_t^{2i+j-2}, \quad (20)$$

where the vector field f_0 is defined by

$$f_0 := \bar{f}_0 + \sum_{i=1}^r (\phi_1 f_i(\phi_2) - \phi_2 f_i(\phi_1)) f_i. \quad (21)$$

The associated infinitesimal generator $\mathcal{L}$ is defined by

$$\mathcal{L}\psi = f_0(\psi) + \frac{1}{2\kappa} \sum_{i=1}^r \sum_{j=1}^2 \phi_j f_i(\phi_j f_i(\psi)), \quad (22)$$

for all $\psi \in C^\infty(\mathcal{M})$. Using the product rule, we compute that

$$\phi_j f_i(\phi_j f_i(\psi)) = \phi_j f_i(\phi_j) f_i(\psi) + \phi_j^2 f_i(f_i(\psi)). \quad (23)$$

Therefore, for any $i \in \{1, \dots, r\}$, we have that

$$\begin{aligned} \sum_{j=1}^2 \phi_j f_i(\phi_j f_i(\psi)) &= (\phi_1^2 + \phi_2^2) f_i(f_i(\psi)) \\ &+ (\phi_1 f_i(\phi_1) + \phi_2 f_i(\phi_2)) f_i(\psi). \end{aligned} \quad (24)$$

From the chain rule and the definition of F_j , we compute that

$$f_i(\phi_1) = \phi_2 f_i(\phi), \quad f_i(\phi_2) = -\phi_1 f_i(\phi), \quad \phi_1^2 + \phi_2^2 = 1.$$

Substituting into (21) and (24), we obtain that

$$f_0 = \bar{f}_0 - \sum_{i=1}^r f_i(\phi) f_i, \quad \sum_{j=1}^2 \phi_j f_i(\phi_j f_i(\psi)) = f_i(f_i(\psi)).$$

Substituting into the infinitesimal generator, we obtain that

$$\mathcal{L}\psi = f_0(\psi) + \frac{1}{2\kappa} \sum_{i=1}^r f_i(f_i(\psi)).$$

If ∇ is any affine connection on $\mathcal{M}$, then, using [27, Proposition 4.15], we compute that

$$f_i(f_i(\psi)) = (\nabla_{f_i} d\psi)(f_i) + \nabla_{f_i} f_i(\psi), \quad (25)$$

In particular, if ∇ is the Levi-Civita connection associated with the metric (10), then

$$\sum_{i=1}^r f_i(f_i(\psi)) = \Delta\psi + \sum_{i=1}^r \nabla_{f_i} f_i(\psi). \quad (26)$$

Therefore, the generator $\mathcal{L}$ acts on a smooth function ψ as

$$\mathcal{L}\psi = f_0(\psi) + \frac{1}{2\kappa} \sum_{i=1}^r \nabla_{f_i} f_i(\psi) + \frac{1}{2\kappa} \Delta\psi, \quad (27)$$

so that, if the vector field $\bar{f}_0$ is taken to be

$$\bar{f}_0 = -\frac{1}{2\kappa} \sum_{i=1}^r \nabla_{f_i} f_i, \quad (28)$$

then the generator $\mathcal{L}$ is given by

$$\mathcal{L}\psi = -\langle \text{grad } \phi, \text{grad } \psi \rangle + \frac{1}{2\kappa} \Delta\psi, \quad (29)$$

for any function $\psi \in C^\infty(\mathcal{M})$. Due to compactness of $\mathcal{M}$ and the fact that the operator $\mathcal{L}$ is uniformly elliptic for any $\kappa > 0$, it follows that its adjoint $\mathcal{L}^*$, defined by

$$\mathcal{L}^*\psi = \text{div}(\psi \text{grad } \phi) + \frac{1}{2\kappa} \Delta\psi,$$

has a unique invariant density. Indeed, the Fokker-Planck equation corresponding to (20) is

$$\begin{aligned} \partial_t \rho &= \mathcal{L}^* \rho = \text{div}(\rho \text{grad } \phi) + \frac{1}{2\kappa} \Delta\rho \\ &= \frac{1}{2\kappa} \text{div}(e^{-2\kappa\phi} \text{grad}(e^{2\kappa\phi} \rho)). \end{aligned}$$

From the last equality, we observe that the density $\rho_\kappa = c_\kappa^{-1} e^{-2\kappa\phi}$, $c_\kappa := \int_{\mathcal{M}} e^{-2\kappa\phi} \text{vol}$, where vol is the Riemannian density, is a stationary solution of the Fokker-Planck operator, and, therefore, $\mu_\kappa := \rho_\kappa \text{vol}$ is the unique invariant measure for (8). ■

Proof of Proposition 1: Fix any open $U \supset S_*$ and define the set $K := \mathcal{M} \setminus U$, which is compact and disjoint from S_* . By continuity of ϕ and compactness, the minimum

$$m_K := \min_{\bar{x} \in K} \phi(\bar{x}),$$

exists and satisfies $m_K > \phi_*$. Choose $\epsilon \in (0, m_K - \phi_*)$ and define the nonempty closed set $N := \{x \in \mathcal{M} : \phi(x) \leq \phi_* + \epsilon\}$. Then, we have that

$$\begin{aligned} \int_K e^{-2\kappa\phi} \text{vol} &\leq e^{-2\kappa m_K} \text{vol}(K), \\ \int_N e^{-2\kappa\phi} \text{vol} &\geq e^{-2\kappa(\phi_* + \epsilon)} \text{vol}(N). \end{aligned}$$

Therefore, because $c_\kappa \geq \int_N e^{-2\kappa\phi} \text{vol}$, we obtain that

$$\mu_\kappa(K) \leq \frac{e^{-2\kappa m_K} \text{vol}(K)}{e^{-2\kappa(\phi_* + \epsilon)} \text{vol}(N)} = C_K e^{-2\kappa c_K},$$

where $C_K := \text{Vol}(K) \text{Vol}(N)^{-1}$ and $c_K := (m_K - \phi_* - \epsilon) > 0$ are independent of κ . Because $c_K > 0$, the right-hand side of the inequality (14) decays exponentially as $\kappa \nearrow \infty$, hence $\mu_\kappa(K) \searrow 0$ and, therefore, $\mu_\kappa(U) \nearrow 1$. ■

Proof of Proposition 2: For any fixed $\kappa > 0$, the operator $\mathcal{L}$ is elliptic with a spectral gap $\lambda_\kappa > 0$. Consequently, the Markov semigroup $e^{t\mathcal{L}}$ satisfies

$$\|e^{t\mathcal{L}} f\|_{L^2(\mu_\kappa)} \leq e^{-\lambda_\kappa t} \|f\|_{L^2(\mu_\kappa)},$$

for any $f \in L^2(\mu_\kappa)$ with $\int_{\mathcal{M}} f \mu_\kappa = 0$. To proceed, define the function

$$u_t := \rho_t \rho_\kappa^{-1} - 1,$$

so that $\rho_t = \rho_\kappa(u_t + 1)$. Since, by definition, $\mathcal{L}^* \rho_\kappa = 0$, and $\mathcal{L}^*(\rho_\kappa v) = \rho_\kappa \mathcal{L}v$ for any smooth v , by self-adjointness of $\mathcal{L}$ in $L^2(\mu_\kappa)$, we obtain that

$$\rho_\kappa \partial_t u_t = \partial_t \rho_t = \mathcal{L}^* \rho_t = \mathcal{L}^*(\rho_\kappa u_t) = \rho_\kappa \mathcal{L}u_t,$$

and hence $\partial_t u_t = \mathcal{L}u_t$. Moreover, direct computation gives

$$\int_{\mathcal{M}} u_t \mu_\kappa = \int_{\mathcal{M}} (\rho_t - \rho_\kappa) \text{vol} = 0,$$

and, consequently, we obtain that

$$\|u_t\|_{L^2(\mu_\kappa)} \leq e^{-\lambda_\kappa t} \|u_0\|_{L^2(\mu_\kappa)},$$

for all $t \geq 0$. From the Cauchy-Schwartz inequality and the positivity of ρ_κ for any $\kappa > 0$, we obtain that

$$\begin{aligned} \|\rho_t - \rho_\kappa\|_{L^1(\text{vol})} &= \int_{\mathcal{M}} |\rho_t - \rho_\kappa| \rho_\kappa^{-\frac{1}{2}} \rho_\kappa^{\frac{1}{2}} \text{vol} \\ &\leq \left(\int_{\mathcal{M}} (\rho_t - \rho_\kappa)^2 \rho_\kappa^{-1} \text{vol} \right)^{\frac{1}{2}} \left(\int_{\mathcal{M}} \rho_\kappa \text{vol} \right)^{\frac{1}{2}} \\ &= \left(\int_{\mathcal{M}} u_t^2 \rho_\kappa \text{vol} \right)^{\frac{1}{2}} = \|u_t\|_{L^2(\mu_\kappa)} \leq e^{-\lambda_\kappa t} \|u_0\|_{L^2(\mu_\kappa)}. \end{aligned}$$

which proves the first claim. To show the second claim, we note that the operator $\mathcal{L}$ belongs to the class of operators considered in [28, Section 8.2]. Therefore, it follows from [28, Proposition 2.1], in conjunction with the assumption that ϕ is a Morse function with a unique local minimum, that the

spectral gap of $\mathcal{L}$ is uniformly bounded below for all $\kappa > 0$ by a strictly positive constant $\lambda > 0$. ■

Proof of Proposition 3: By definition, the total variation distance between the probability laws $\mathbb{P}$ and $\mathbb{Q}$ is

$$\sup_{A \text{ measurable}} |\mathbb{P}(A) - \mathbb{Q}(A)|. \quad (30)$$

When the laws $\mathbb{P}$ and $\mathbb{Q}$ have densities p and q , it follows that

$$\sup_{A \text{ measurable}} |\mathbb{P}(A) - \mathbb{Q}(A)| = \frac{1}{2} \|p - q\|_{L^1(\text{vol})}. \quad (31)$$

Substituting for $\mathbb{P}$ with the law of the random variable $\bar{x}_t$ and for $\mathbb{Q}$ with the invariant measure μ_κ , we obtain that

$$|\mathbb{P}\{\bar{x}_t \in U\} - \mu_\kappa(U)| \leq \frac{1}{2} \|\rho_t - \rho_\kappa\|_{L^1(\text{vol})}, \quad (32)$$

where ρ_t solve the Fokker-Planck equation from the initial condition ρ_0 . Hence, we obtain that

$$|\mathbb{P}\{\bar{x}_t \in U\} - \mu_\kappa(U)| \leq \frac{1}{2} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})} e^{-\lambda_\kappa t}, \quad (33)$$

for any measurable $U \subset \mathcal{M}$, any $\kappa > 0$, and any $t \geq 0$. If, in addition, ρ_0 is the uniform probability density, then

$$\begin{aligned} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})}^2 &= \int_{\mathcal{M}} (\rho_0 - \rho_\kappa)^2 \rho_\kappa^{-1} \text{vol} \\ &= \int_{\mathcal{M}} \rho_0^2 \rho_\kappa^{-1} \text{vol} - 2 \int_{\mathcal{M}} \rho_0 \text{vol} + \int_{\mathcal{M}} \rho_\kappa \text{vol} \\ &= \frac{1}{\text{Vol}(\mathcal{M})^2} \int_{\mathcal{M}} \rho_\kappa^{-1} \text{vol} - 1, \end{aligned}$$

where we used the fact that $\rho_0 = \text{Vol}(\mathcal{M})^{-1}$, $\int_{\mathcal{M}} \rho_\kappa \text{vol} = 1$, and $\int_{\mathcal{M}} \rho_0 \text{vol} = 1$. Recalling the form of the invariant measure, we see that

$$\rho_\kappa(x) \geq \frac{e^{-2\kappa \max_{x \in \mathcal{M}} \phi(x)}}{e^{-2\kappa \min_{x \in \mathcal{M}} \phi(x)} \int_{\mathcal{M}} \text{vol}} \geq \frac{e^{-2\gamma_\phi \kappa}}{\text{Vol}(\mathcal{M})}, \quad (34)$$

for all $x \in \mathcal{M}$. Therefore, we obtain the bound

$$\begin{aligned} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})}^2 &\leq \frac{e^{2\gamma_\phi \kappa}}{\text{Vol}(\mathcal{M})} \int_{\mathcal{M}} \frac{1}{\rho_\kappa} \text{vol} - 1 \\ &= e^{2\gamma_\phi \kappa} - 1 \leq e^{2\gamma_\phi \kappa}, \end{aligned}$$

from which the second inequality follows. ■

Proof of Proposition 4: Using the triangle inequality, we have that

$$\begin{aligned} |\mathbb{P}\{x_t^\varepsilon \in U\} - \mu_\kappa(U)| &\leq |\mathbb{P}\{x_t^\varepsilon \in U\} - \mathbb{P}\{\bar{x}_t \in U\}| + |\mathbb{P}\{\bar{x}_t \in U\} - \mu_\kappa(U)| \\ &\leq |\mathbb{P}\{x_t^\varepsilon \in U\} - \mathbb{P}\{\bar{x}_t \in U\}| + \frac{1}{2} \|\rho_0 - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})} e^{-\lambda_\kappa t} \end{aligned}$$

where we use Corollary 3. Since $\rho_0 \in C^\infty(\mathcal{M})$, the law of $\bar{x}_t$ admits a density for all $t \geq 0$ and, thus, any open set $U \subset \mathcal{M}$ with a smooth boundary is a continuity set of the law of $\bar{x}_t$. Consequently, from Theorem 1, we obtain that there exists $\varepsilon^* > 0$ such that, for all $\varepsilon \in (0, \varepsilon^*)$ and all $t \in [0, T]$,

$$|\mathbb{P}\{x_t^\varepsilon \in U\} - \mathbb{P}\{\bar{x}_t \in U\}| < \varepsilon, \quad (35)$$

which concludes the proof. ■

Proof of Lemma 1: On $[k\varepsilon, (k+1)\varepsilon)$, differentiating (6) in t eliminates the additive constant $W_{k\varepsilon}^\ell$, so $b_\varepsilon^\ell(t, W)$ depends

on W only through the increments w_k which, by definition, are i.i.d samples from the Gaussian distribution $\mathcal{N}(0, \varepsilon I)$. Therefore, for any fixed $w \in \mathbb{R}^{2r}$, if we define $\Psi_\varepsilon(x, w)$ as the time-1 solution of the ODE

$$y'(s) = \varepsilon \bar{f}_0(y) + \frac{1}{\sqrt{\kappa}} \sum_{i=1}^r \sum_{j=1}^2 g_{i,j}(y) w^{2i+j-2} (U_\kappa^j(w))'(s), \quad (36)$$

from the initial condition $y(0) = x$, where the vector fields $g_{i,j}$ are defined by $g_{i,j} := F_j \circ \phi f_i$, we see that $\Psi_\varepsilon(x, w)$ is independent of k , and, therefore, the sampled process X_k is a time-homogeneous Markov process, where the Markov property follows from the independence of the increments w_k . To complete the proof, it remains to show that P_ε maps $C(\mathcal{M})$ into $C(\mathcal{M})$. Since $w^{2i-1} w^{2i} = 0$ if and only if one of the two coordinates vanishes, the set $\mathcal{D}_0 := \bigcup_i \{w \in \mathbb{R}^{2r} : w^{2i-1} w^{2i} = 0\}$ coincides with the union of the coordinate hyperplanes $\{w \in \mathbb{R}^{2r} \mid w^\ell = 0\}$. In particular, $\mathcal{D}_0$ is closed and $\text{Leb}(\mathcal{D}_0) = 0$, which implies that it also has zero measure with respect to the Gaussian measure $\mathcal{N}(0, \varepsilon I)$ since the latter is absolutely continuous with respect to the former. Its complement is the disjoint union of the 4^r open orthants of $\mathbb{R}^{2r}$. On each orthant, the sign pattern of w , and therefore the branch $U_i^j(w)$ of every pair in (36) is constant. Consequently, for w ranging over a fixed orthant, the right-hand side of (36) is a single smooth function of (y, w, s) . By smooth dependence of ODE solutions on initial conditions and parameters, the map Ψ_ε is jointly smooth on $\mathcal{M} \times O$ for each orthant $O \subset \mathbb{R}^{2r}$. Borel measurability of Ψ_ε on all of $\mathcal{M} \times \mathbb{R}^{2r}$ follows, since its domain partitions into finitely many Borel sets, on each of which it is continuous.

Nevertheless, continuity of $\Psi_\varepsilon(x, \cdot)$ on all of $\mathbb{R}^{2r}$ fails. Indeed, the map $\Psi_\varepsilon(x, \cdot)$ has a genuine discontinuity across the set $\mathcal{D}_0$. However, because the discontinuity set $\mathcal{D}_0$ has zero measure, we are allowed to proceed as follows: Fix a function $\psi \in C(\mathcal{M})$, a sequence $x_n \rightarrow x$ in $\mathcal{M}$, and define

$$h_n(w) := \psi(\Psi_\varepsilon(x_n, w)), \quad h(w) := \psi(\Psi_\varepsilon(x, w)).$$

For every $w \notin \mathcal{D}_0$, the point w lies in some orthant O , and $(x_n, w) \rightarrow (x, w)$ within $\mathcal{M} \times O$, on which Ψ_ε is continuous. Hence, $h_n(w) \rightarrow h(w)$, and $h_n \rightarrow h$ pointwise $\mathcal{N}(0, \varepsilon I)$ -almost everywhere. Since, by definition, $|h_n| \leq \|\psi\|_\infty$, the Dominated Convergence theorem [29, Theorem 3.3.1] yields

$$P_\varepsilon \psi(x_n) = \int h_n d\gamma_\varepsilon \rightarrow \int h d\gamma_\varepsilon = P_\varepsilon \psi(x),$$

i.e., $P_\varepsilon \psi$ is sequentially continuous whenever $\psi \in C(\mathcal{M})$. Because $\mathcal{M}$ is metrizable, sequential continuity implies continuity, and we have $P_\varepsilon \psi \in C(\mathcal{M})$. ■

Proof of Proposition 5: From [26, Prop. 6.1.1], the chain (X_k) is weak Feller since its kernel P_ε satisfies $P_\varepsilon \psi \in C(\mathcal{M})$ for any $\psi \in C(\mathcal{M})$. From [26, Prop. 12.0.1], it follows that the chain (X_k) admits at least one invariant measure. ■

Proof of Theorem 3: Since (8) is uniformly elliptic with smooth coefficients on the compact $\mathcal{M}$, its transition kernel admits a density $p_t(x, y)$, jointly smooth and strictly positive for $t > 0$. Moreover, for any $\kappa > 0$, the corresponding operator

$\mathcal{L}$ has a strictly positive spectral gap $\lambda_\kappa > 0$. Define the constant

$$C_\kappa := \sup_{x \in \mathcal{M}} \|p_1(x, \cdot) - \rho_\kappa\|_{L^2(\rho_\kappa^{-1} \text{vol})},$$

which is finite by joint continuity of p_1 on $\mathcal{M} \times \mathcal{M}$ and the positivity of ρ_κ . For $t \geq 1$, the Markov property identifies the density ρ_t^x of $\bar{x}_t(x)$ with the solution of the Fokker–Planck equation at time $t - 1$ from the initial density $p_1(x, \cdot) \in C^\infty(\mathcal{M})$, so Proposition 2 yields

$$\|\rho_t^x - \rho_\kappa\|_{L^1(\text{vol})} \leq C_\kappa e^{-\lambda_\kappa(t-1)},$$

uniformly in x . Let $T := 1 + \lambda_\kappa^{-1} \log(2C_\kappa \delta^{-1})$. Then, for every $t \geq T$ and every f with $\|f\|_\infty \leq 1$,

$$\sup_{x \in \mathcal{M}} |\mathbb{E}f(\bar{x}_t(x)) - \mu_\kappa(f)| \leq \sup_{x \in \mathcal{M}} \|\rho_t^x - \rho_\kappa\|_{L^1(\text{vol})} \leq \frac{\delta}{2}.$$

Next, let $\varepsilon^* := \varepsilon^*(\frac{\delta}{2}, T+1)$ be given by Theorem 1, fix $\varepsilon \in (0, \varepsilon^*]$, and set $k_\varepsilon := \lceil T\varepsilon^{-1} \rceil$, so that $t_\varepsilon := k_\varepsilon \varepsilon \in [T, T+1]$ and $P_\varepsilon^{k_\varepsilon}(x, \cdot)$ is the law of $x_{t_\varepsilon}^\varepsilon(x)$. Then, the convergence of trajectories property (9) along with Jensen's inequality give

$$|P_\varepsilon^{k_\varepsilon} f(x) - \mathbb{E}f(\bar{x}_{t_\varepsilon}(x))| \leq \mathbb{E}|x_{t_\varepsilon}^\varepsilon(x) - \bar{x}_{t_\varepsilon}(x)| \leq \frac{\delta}{2},$$

for f with $\|f\|_\infty \leq 1$, $\text{Lip}(f) \leq 1$, uniformly in x , so that

$$\sup_x d_{\text{BL}}(P_\varepsilon^{k_\varepsilon}(x, \cdot), \mu_\kappa) \leq \delta \quad (37)$$

Next, let $k \in \mathbb{N}$ with $k \geq T\varepsilon^{-1}$, so that $k \geq k_\varepsilon$. Because $P_\varepsilon^{k-k_\varepsilon}(x, \cdot)$ is a probability measure, the Chapman–Kolmogorov relation allows us to write

$$P_\varepsilon^k f(x) - \mu_\kappa(f) = \int_{\mathcal{M}} (P_\varepsilon^{k_\varepsilon} f(y) - \mu_\kappa(f)) P_\varepsilon^{k-k_\varepsilon}(x, dy),$$

and, since the bound (37) is uniform in the base point, the right-hand side is bounded in absolute value by δ . Taking the supremum over all f with $\|f\|_\infty \leq 1$ and $\text{Lip}(f) \leq 1$ yields the assertion. Finally, if $\mu_\varepsilon P_\varepsilon = \mu_\varepsilon$, then $\mu_\varepsilon = \mu_\varepsilon P_\varepsilon^{k_\varepsilon}$, so that

$$\mu_\varepsilon(f) - \mu_\kappa(f) = \int_{\mathcal{M}} (P_\varepsilon^{k_\varepsilon} f(x) - \mu_\kappa(f)) \mu_\varepsilon(dx),$$

for any such f , which is again bounded by δ in absolute value, and, therefore, $d_{\text{BL}}(\mu_\varepsilon, \mu_\kappa) \leq \delta$. ■

Proof of Lemma 2: We first claim that, for every Borel set $A \subseteq \mathcal{M}$ and every $\eta \in (0, 1]$, we have that

$$\inf_{x \in \mathcal{M}} P_\varepsilon^k(x, A^\eta) \geq \mu_\kappa(A) - \delta \eta^{-1}, \quad (38)$$

$$\sup_{x \in \mathcal{M}} P_\varepsilon^k(x, A) \leq \mu_\kappa(A^\eta) + \delta \eta^{-1}. \quad (39)$$

Indeed, the function $f := \max\{0, 1 - \eta^{-1}d(\cdot, A)\}$ satisfies

$$\mathbf{1}_A \leq f \leq \mathbf{1}_{A^\eta}, \quad \|f\|_\infty \leq 1, \quad \text{Lip}(f) \leq \eta^{-1}.$$

Since $\eta \leq 1$, the rescaled function ηf satisfies $\|\eta f\|_\infty \leq 1$ and $\text{Lip}(\eta f) \leq 1$, so applying Theorem 3 to ηf and dividing by η , using linearity of P_ε^k and of μ_κ , yields

$$|P_\varepsilon^k f(x) - \mu_\kappa(f)| \leq \delta \eta^{-1}, \quad \forall x \in \mathcal{M}.$$

The claim then follows from the two chains of inequalities

$$P_\varepsilon^k(x, A^\eta) \geq P_\varepsilon^k f(x) \geq \mu_\kappa(f) - \delta \eta^{-1} \geq \mu_\kappa(A) - \delta \eta^{-1},$$

$$P_\varepsilon^k(x, A) \leq P_\varepsilon^k f(x) \leq \mu_\kappa(f) + \delta \eta^{-1} \leq \mu_\kappa(A^\eta) + \delta \eta^{-1}.$$

To prove the first claim, note that $\eta_U > 0$, since $S_\star$ is compact, $\mathcal{M} \setminus U$ is closed and nonempty, and the two sets are disjoint. We now apply (38) with $A = S_\star^{\eta_U}$ and $\eta := \eta_U$. By the triangle inequality, we have that $(S_\star^{\eta_U})^{\eta_U} \subseteq S_\star^{2\eta_U}$. Also, since $2\eta_U \leq d(S_\star, \mathcal{M} \setminus U)$, we have that $S_\star^{2\eta_U} \subseteq U$. It follows that

$$P_\varepsilon^k(x, U) \geq P_\varepsilon^k(x, (S_\star^{\eta_U})^{\eta_U}) \geq \mu_\kappa(S_\star^{\eta_U}) - \delta \eta_U^{-1},$$

and, since the set $\mathcal{M} \setminus S_\star^{\eta_U}$ is closed and disjoint from $S_\star$, we obtain from (14) that

$$\mu_\kappa(S_\star^{\eta_U}) \geq 1 - C_U e^{-2\kappa c_U},$$

which proves the first claim.

To prove the second claim, note that $\eta_K > 0$, since K is compact, $S_\star$ is closed, and the two sets are disjoint. We apply (39) with $A := K$ and $\eta := \eta_K$, so that

$$P_\varepsilon^k(x, K) \leq \mu_\kappa(K^{\eta_K}) + \delta \eta_K^{-1} \leq \mu_\kappa(\overline{K^{\eta_K}}) + \delta \eta_K^{-1}.$$

By definition, every $y \in \overline{K^{\eta_K}}$ satisfies $d(y, K) \leq \eta_K$, while, by construction $d(K, S_\star) \geq 2\eta_K$. Hence, the set $\overline{K^{\eta_K}}$ is closed and disjoint from $S_\star$. We then obtain from (14) that

$$\mu_\kappa(\overline{K^{\eta_K}}) \leq C_K e^{-2\kappa c_K},$$

which proves the second claim and concludes the proof. ■

VII. NUMERICAL EXAMPLE

Let $\mathcal{M} = \mathbb{S}^2$ and $\mathcal{F} = \{f_1, f_2, f_3\}$ where the vector fields f_i are given by $f_1(x) = x^2 \partial_1 - x^1 \partial_2$, $f_2(x) = x^3 \partial_2 - x^2 \partial_3$, and $f_3(x) = x^1 \partial_3 - x^3 \partial_1$ in the ambient coordinates on $\mathbb{R}^3$. Fix any $x_\star \in \mathbb{S}^2$ and let ϕ be given by

$$\phi(x) := 1 - (x_\star^1 x^1 + x_\star^2 x^2 + x_\star^3 x^3).$$

The function ϕ is a Morse function. Moreover, ϕ has two critical points, namely, $\pm x_\star$, and only the critical point $x_\star$ is a local minimum. The Riemannian metric $\langle \cdot, \cdot \rangle$ corresponding to the family $\mathcal{F}$ is defined in the ambient coordinates by

$$\langle \cdot, \cdot \rangle = \sum_{i,j=1}^3 g_{ij} dx^i \otimes dx^j, \quad G(x) = [g_{ij}] = I - xx^\top. \quad (40)$$

In particular, the metric $\langle \cdot, \cdot \rangle$ coincides with the induced metric inherited from the ambient Euclidean metric on $\mathbb{R}^3$. Consequently, if $D_X Y$ denotes the directional derivative of the vector field Y along the vector field X in the ambient space $\mathbb{R}^3$, then the Levi-Civita connection on $\mathcal{M}$ in the ambient coordinates is given by $\nabla_X Y = G(x)^\dagger D_X Y$, where $G(x)^\dagger$ is the Moore–Penrose inverse of $G(x)$. Utilizing this expression, we compute that $\tilde{f}_0 = 0$. We now generate several sample paths of the random ODE (7) from the initial condition $-x_\star$. For simplicity, we fix $x_\star = (0, 0, 1)$. A plot of some sample paths of the chain X_k on $\mathcal{M}$ are shown in Figure 1 for $\varepsilon = 0.02$, and $\kappa = 25$. As expected, the sample paths converge to the vicinity global minimizer of the function ϕ . Moreover, the trajectories remain on $\mathcal{M}$ for all time. We also plot ensemble statistics by simulating $N = 256$ independent sample paths and computing the empirical approximations for the probability $\mathbb{P}\{X_k \in U\}$, where $U := \{x \in \mathcal{M} \mid \phi(x) < 0.06\}$,

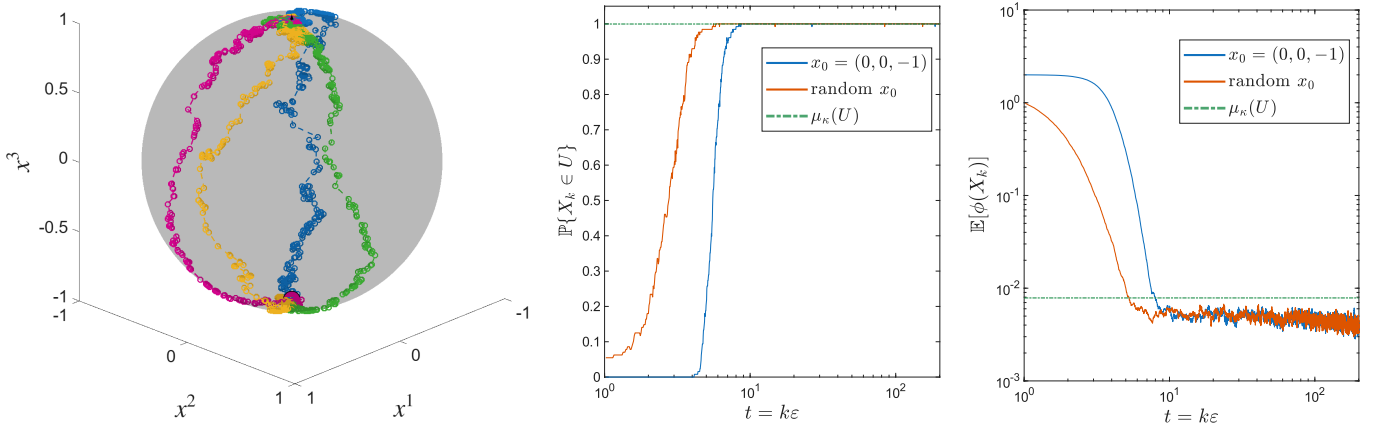


Fig. 1: (Left) the sample paths generated by the random ODE (7), (center) empirical approximations of $\mathbb{P}\{X_k \in U\}$, and (right) empirical approximations of $\mathbb{E}[\phi(X_k)]$.

as well as the expectation $\mathbb{E}[\phi(X_k)]$. As shown in Figure 1, the approximations converge to reasonable margins of the corresponding values for the associated Langevin diffusion, which corroborates our analysis.

VIII. CONCLUSIONS

We introduced and analyzed a novel stochastic Lie-bracket approximation algorithm for global zeroth-order optimization on compact manifolds without boundary. Our analysis shows that the proposed algorithm leads to finite-horizon, as well as infinite-horizon, practical probabilistic guarantees on the convergence of the proposed scheme. Numerical results are provided to illustrate the behavior of the algorithm. Future work will consider extensions of the proposed algorithm to compact manifolds with boundaries.

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